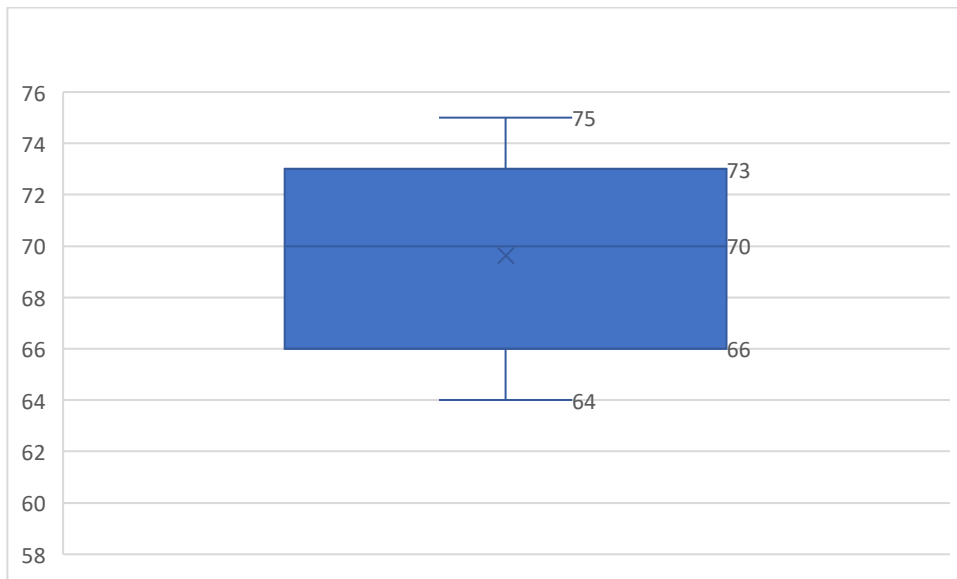


SOLUTIONS FOR MOCK EXAM – 2024-2025

Solution 1

- a) The Five Number Summary is a set of the five most important sample percentiles:
- the sample minimum (smallest observation) is 64
 - the lower quartile or first quartile (Q1): $(n+1)/4^{\text{th}} = 3^{\text{rd}}$ datapoint is 66
 - the median (the middle value): $(n+1)/2^{\text{th}} = 6^{\text{th}}$ observation is 70
 - the upper quartile or third quartile (Q3): $3(n+1)/4^{\text{th}} = 9^{\text{th}}$ datapoint is 73
 - the sample maximum (largest observation) is 75.

b)



c) The interquartile range of the data equals to $IQR = Q3 - Q1 = 73 - 66 = 7$

d) The mean value of heights is $\mu = 69.64$. The standard deviation of heights is $\sigma =$

$$\sqrt{\frac{\sum_i (x_i - \mu)^2}{n}} = 3.42$$

Height	Mean	$(x_i - \mu)^2$
64	69.64	31.77
65	69.64	21.50

66	69.64	13.22
69	69.64	0.40
69	69.64	0.40
70	69.64	0.13
70	69.64	0.13
71	69.64	1.86
73	69.64	11.31
74	69.64	19.04
75	69.64	28.77

$$\begin{array}{rcl}
 \sum (x_i - \mu)^2 & & 128.55 \\
 (\sum (x_i - \mu)^2) / n & & 11.69 \\
 \Sigma & & \mathbf{3.42}
 \end{array}$$

Standard deviation:

$$\sigma = \sqrt{(\sum (x - \mu)^2) / n}$$

$$\mu = 64 + 65 + 66 + 69 + 69 + 70 + 70 + 71 + 73 + 74 + 75 / 11$$

$$= 69.6$$

$$\sum (x - \mu)^2 = (64 - 69.6)^2 + (65 - 69.6)^2 + (66 - 69.6)^2 + (69 - 69.6)^2 + (69 - 69.6)^2 + (70 - 69.6)^2 + (70 - 69.6)^2 + (71 - 69.6)^2 + (73 - 69.6)^2 + (74 - 69.6)^2 + (75 - 69.6)^2 = 128.56$$

$$\sigma = \sqrt{(128.56 / 11)}$$

$$\sigma = 3.419$$

Solution 2

- a) $P(\text{battery is defective}) = (20+10)/1000 = 0.03$
- b) $P(\text{battery is working}) = 1 - 0.03 = 0.97$
- c) $P(\text{battery is defective} \mid \text{selected battery is AA}) = 20/(480+20) = 0.04$

a)

The probability of an event is calculated by dividing the number of favourable outcomes by the total number of possible outcomes.

1. Total number of batteries is $20 + 480 + 10 + 490 = 1000$ and the number of defective batteries is $20 + 10 = 30$.
2. Probability that a randomly selected battery is defective is $30 / 1000 = 0.03$.

This means that if you randomly select a single battery out of the box, there is a **3%** chance that it will be defective.

b)

The probability of an event is calculated by dividing the number of favourable outcomes by the total number of possible outcomes.

1. Total number of batteries is $20 + 480 + 10 + 490 = 1000$ and the number of working batteries is $480 + 490 = 970$.
2. The probability that a randomly selected battery is working is $970 / 1000 = 0.97$.

This means that if you randomly select a single battery out of the box, there is a **97%** chance that it will be working.

c)

The probability of an event A given that another event B has occurred is calculated by dividing the probability of both events A and B occurring by the probability of event B. Event A is that the battery is defective and event B is that the battery is of AA type.

The probability of a battery being defective and of AA type is calculated by dividing the number of defective AA batteries by the total number of batteries: $20 / 1000 = 0.02$.

The probability of a battery being of AA type is calculated by dividing the total number of AA batteries by the total number of batteries: $(20 + 480) / 1000 = 0.5$.

So the probability that a battery is defective given that it is of AA type is $0.02 / 0.5 = 0.04$.

This means that if you know that a randomly selected battery is of AA type, there is a **4%** chance that it will be defective.

Solution 3

a)

$P(r \text{ successes in } n \text{ goes}) = {}^nC_r p^r (1 - p)^{(n-r)}$, where

n is the number of goes,

r is the number of required “successes”, and

p is the probability of the result you’re looking for.

p 48%

n 10

r 5

$P(5 \text{ people having a blood type O in a group of } 10) = 0.24$

b)

$P(r \text{ successes in } n \text{ goes}) = {}^nC_r p^r (1 - p)^{(n-r)}$ where

n is the number of goes,

r is the number of required “successes”, and

p is the probability of the result you’re looking for

p 48%

n 5

r 0

$P(0 \text{ people having a blood type O in a group of } 5) = 0.04$

$P(\text{of at least one person having blood type O in a group of } 5) = 1 - 0.04 = 0.96$

Solution 4

$$z = (x - \mu)/\sigma$$

Millie's mark:

$$Z = (80 - 65)/12 = 1.25$$

Rose's mark:

$$Z = (680 - 600)/100 = 0.8$$

Millie's mark is 1.25 standard deviations above mean, and Rose's mark is only 0.8 standard deviations above mean, therefore it is Millie who did better on the exam.

Solution 5

a)

Mean = 0.51 kg, standard deviation = 0.05 kg

b)

$$\text{SEM} = \frac{s}{\sqrt{n}} = \frac{0.05}{\sqrt{10}} = 0.02$$

$$t = 2.262$$

95% confidence interval: $0.51 \pm 2.262 \times 0.02$
= 0.47kg to 0.56 kg

c)

$$t = 3.250$$

99% confidence interval: $0.51 \pm 3.250 \times 0.02$
= 0.455 kg to 0.575 kg

Solution 6

To find the expected return, we multiply each potential return by its corresponding probability and then sum up the results.

Expected return = (Return of Strategy A * Probability of Strategy A) + (Return of Strategy B * Probability of Strategy B) + (Return of Strategy C * Probability of Strategy C)

$$\begin{aligned}\text{Expected return} &= (£500 * 0.25) + (£300 * 0.40) + (£200 * 0.35) \\ &= £125 + £120 + £70 \\ &= £315\end{aligned}$$

Therefore, the expected return of the marketing investment is £315.

Solution 7

Let's use Bayes' theorem to solve the proposed question:

Given Information:

- Probability of a transaction being fraudulent: $P(\text{Fraudulent}) = 0.03$
- Probability that the algorithm incorrectly flags a legitimate transaction as fraudulent: $P(\text{Algorithm Flags Fraud} \mid \text{Not Fraudulent}) = 0.08$
- Probability that the algorithm correctly identifies a fraudulent transaction: $P(\text{Algorithm Flags Fraud} \mid \text{Fraudulent}) = 0.95$

Required:

Calculate the probability that a flagged transaction is not fraudulent, given that the algorithm has identified it as fraudulent:

$$P(\text{Not Fraudulent} \mid \text{Algorithm Flags Fraud})$$

Bayes' Theorem:

$P(\text{Not Fraudulent} \mid \text{Algorithm Flags Fraud}) =$

$$\frac{P(\text{Not Fraudulent}) * P(\text{Algorithm Flags Fraud} \mid \text{Not Fraudulent})}{P(\text{Algorithm Flags Fraud})}$$

Calculation:

$$\begin{aligned} P(\text{Algorithm Flags Fraud}) \\ &= P(\text{Not Fraudulent}) \times P(\text{Algorithm Flags Fraud} \mid \text{Not Fraudulent}) \\ &+ P(\text{Fraudulent}) \times P(\text{Algorithm Flags Fraud} \mid \text{Fraudulent}) \end{aligned}$$

$$\begin{aligned} P(\text{Algorithm Flags Fraud}) \\ &= (1 - P(\text{Fraudulent})) \times 0.08 + P(\text{Fraudulent}) \times 0.95 \end{aligned}$$

$$P(\text{Algorithm Flags Fraud}) = (1 - 0.03) \times 0.08 + 0.03 \times 0.95$$

$$P(\text{Algorithm Flags Fraud}) = 0.97 \times 0.08 + 0.03 \times 0.95$$

$$P(\text{Algorithm Flags Fraud}) = 0.0776 + 0.0285$$

$$P(\text{Algorithm Flags Fraud}) = 0.1061$$

Now, we can use Bayes' theorem:

$$P(\text{Not Fraudulent} | \text{Algorithm Flags Fraud}) = \frac{(1 - P(\text{Fraudulent})) \times P(\text{Algorithm Flags Fraud} | \text{Not Fraudulent})}{P(\text{Algorithm Flags Fraud})}$$

$$P(\text{Not Fraudulent} | \text{Algorithm Flags Fraud}) = \frac{(1 - 0.03) \times 0.08}{0.1061}$$

$$P(\text{Not Fraudulent} | \text{Algorithm Flags Fraud}) = \frac{0.97 \times 0.08}{0.1061}$$

$$P(\text{Not Fraudulent} | \text{Algorithm Flags Fraud}) = \frac{0.0776}{0.1061}$$

$$P(\text{Not Fraudulent} | \text{Algorithm Flags Fraud}) \approx 0.731$$

Therefore, the corrected probability that a flagged transaction is not fraudulent, given that the algorithm has identified it as fraudulent, is approximately 0.731 or 73.1%.